Sathya Devarajan

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Education

Purdue University | John Martinson Honors College, B.S. in **Mechanical Engineering** | **GPA: 3.8** | Expected May 2028

Experience

Lockheed Martin Sikorsky — Stratford, CT
Tooling Design Engineer Intern — *June 2026 – Present*

- Designed and revised 14 production tooling solutions for CH-53K, Black Hawk, and search-and-rescue rotorcraft components, tailoring designs and tolerancing to fit each part's production method (3-axis/5-axis CNC, manual machining, or additive manufacturing); 3 designs have since been produced and put into use
- Created bills of materials (BOMs) and managed tooling designs through PLM software throughout the design and revision process
- Modified 2 of the 14 tool designs to correct functional issues and reduce manufacturing lead times
- Co-designed a hybrid-propulsion system for a VTOL aircraft in a cross-functional intern competition, integrating engine, motor, battery, and gearbox subsystems within weight and thermal limits
- Presented final design work to engineering leadership and program organizers

Purdue Electric Racing (FSAE Electric) — West Lafayette, IN
Drivetrain Engineer — *August 2025 – Present*

- Designed rear upright & planet carrier for new outrunner drivetrain architecture Siemens NX
- Drafted detailed technical drawings of rear uprights in Siemens NX for internal bearing and seal bores, ensuring precise tolerancing to improve the accuracy of manual boring operations
- Optimized wheel upright manufacturing by modeling toolpaths for 5-axis CNC milling, **reducing machining time** by **50%** compared to previous 3-axis operations while maintaining strict geometric tolerances per GD&T standards
- Developed and fabricated upright fixture plate for 3-axis CNC operation

Projects

Custom Electric Bike Design — *October 2025 – Present*

- Designed a parametric frame model in Siemens NX, enabling rapid modification of critical dimensions like wheelbase and steerer angle to accelerate design iteration
- Validated structural integrity via Ansys FEA, iterating frame dimensions to minimize mass while maintaining a **minimum safety factor** of **2.0** under static load cases

Autonomous Supply Rover — *February 2026 – April 2026*

- Assembled a compact drivetrain with a cargo system utilizing a custom worm-gear mechanism to **transport 500g loads up 35° inclines**
- Programmed a Python closed-loop control system, utilizing real-time IMU feedback to dynamically adjust motor PWM, ensuring precise line tracking over rough terrain

Analog Audio Equalizer — *April 2026*

- Designed schematic for audio equalizer and amplifier to recombine adjusted treble, mids, and bass frequencies for customizable music output
- Troubleshooted and validated breadboard circuit assembly via frequency generator, oscilloscope, and multimeter

Complex Surface Topography — *December 2025 – January 2026*

- Built 3D mesh geometry from 2D heightmaps in Fusion 360 to model complex organic topology while ensuring manufacturability
- Accelerated CAM toolpaths, **cutting machining time** by **60%** while retaining high-level detail without manual post-processing

Smog Tower Simulation — *March 2025*

- Developed Python simulation and conducted financial study on costs and benefits... etc

Leadership

Teen Philanthropy Initiative — Naperville, IL
Community Student Leader — *September 2023 – April 2025*

- Spearheaded a fundraising campaign **securing $15,000 in capital** by authoring successful grant proposals and coordinating direct community outreach appeals
- **Managed** the **strategic allocation** of **$10,000** in grants by conducting site audits and executive interviews to evaluate the financial viability and operational impact of nonprofit applicants

SKILLS

CAD/CAM: CATIA, Siemens NX, Fusion 360, Creo, DS 3DExperience CAD & PLM, Siemens Teamcenter

Manufacturing: 5-Axis & 3-Axis CNC, Manual Machining, Soldering

Analysis: Ansys FEA, GD&T, Dive CFD

Controls/Electronics: Python, MATLAB, Arduino, LTSpice, KiCad